

ELIJAH IFECHUKWU NELSON

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EDUCATION

University of Cambridge, M.Phil. in Machine Learning and Machine Intelligence Cambridge, UK
(Advanced Probabilistic ML, Optimization & Reinforcement Learning) Oct 2025 – Aug 2026

Covenant University (CU), B.Eng. in Electrical and Electronics Engineering Ogun, Nigeria
Grade: First Class Honors; **CGPA:** 4.98/5.0 (Highest in Department's History) Aug 2018 – Sept 2023
Class rank: 1 of 132; **Institute rank:** 1 of 1,175 (Valedictorian)

Thesis: Design, simulation, and implementation of a multi-purpose unmanned aerial vehicle (UAV) with autonomous navigation, precision landing, and visual tracking capabilities [\[LINK\]](#)

RESEARCH INTERESTS

ML + Systems, Embedded Systems, Embedded AI, Sensor Design, IoT, Embedded Software & OS, Computer Architecture, Robotics, Integrated Circuits

RESEARCH & TEACHING EXPERIENCE

AI Research Assistant, Intelligent Manipulation Lab, University of Lincoln, Lincoln, UK Mar 2022 – June 2023

- Conducted experiments on developing AI ensemble models and algorithms and modifying tactile data representations to advance slip classification in robotic manipulation [\[LINK\]](#)
- Developed a novel deep-learning technique, SMOTE-CNN, along with other methods, to achieve accurate predictions on a limited and imbalanced dataset of temporal slip states without relying on its kinematic state, resulting in a 98% slip detection accuracy

Research Assistant, Dept. of Electrical & Electronics Engineering, CU, Ogun, Nigeria Jan 2023 – July 2023

- Designed and developed several projects on embedded software for autonomous and IoT systems (e.g., automated waste management and autonomous navigation for a vacuum cleaning robot)
- Taught and trained over 50 students on research methodologies in embedded systems development

Tutor, Covenant University, Ogun, Nigeria Jan 2021 – July 2023

- Taught and provided personally curated notes and interactive case studies to more than 200 students in Electromagnetism, Applied Electronics, Advanced Control Systems (ACS), Power Electronics, and High Voltage Engineering, leading to an average pass rate of 85% across all subjects, with a 77% pass rate in ACS

PUBLICATIONS, PEER-REVIEWED

- T. Somefun, **N. Elijah**, A. E. Igbo-Orere, O. Timilehin, C. Somefun, and S. Ongbali, "Energy Optimization Algorithm for Reducing Energy Consumption in a Smart Home," 2023 2nd International Conference on Multidisciplinary Engineering and Applied Science, pp. 1-5, doi:10.1109/ICMEAS58693.2023.1037932 [\[LINK\]](#)
- **N. Elijah**, K. Nazari, A. Ghalamzad, "Advancing Slip Classification in Robotic Manipulation through Tactile Data Representation and Model Selection," ICCMA (accepted for publication) [\[LINK\]](#)

PROFESSIONAL EXPERIENCE

Systems Engineer (Founding), Climate in Africa, Lagos, Nigeria Oct 2023 – Aug 2025

- Developed low-cost, WMO-compliant Modular In situ Reporting Instruments (MIRI), with 14 meteorological parameters, in a mission to deploy over **10,000** units across Africa in the next 3 years, with a presence in Nigeria, Ivory Coast, Kenya & Ghana in the first 3 months

- Developed a NOR Flash (MT25QL01GBBB) driver & co-designed CRC-verified **OTA** updates over **BLE** & octet-streams, for resilient field updates; ported 3K LOC from ATmega to STM32 (F4/F103) resulting in 30% increase in performance
- Designed firmware for MIRI (3.5 days per charge) using a sleep scheduler, **RTOS**, and **non-blocking** state machines to reduce GSM/GPRS latency by ~70%, with autonomous fallback logging to NOR Flash/SD card in case of connectivity loss
- Developed a non-blocking task scheduler firmware for autonomous satellite tracking (using azimuth and elevation actuation) & a personally designed **LoRa**-based base-to-ground-station for telemetry and control
- Developed the mail and notification architecture of climateinafrica.com that aggregates TBs of data from all MIRI climate monitors & users; architected a NOAA pipeline to scrape and parse GB-scale climate datasets for our analytics sandbox
- Led the entire product documentation for climateinafrica.com

Machine Learning Engineer, [Ticki](#), Lagos, Nigeria

Aug 2024 – July 2025

- Architected a scalable facial-similarity retrieval system, using OpenCV, FAISS, and async Celery tasks, which optimized end-to-end latency by **70% (sub-100 ms)** through in-memory caching, and concurrent feature search across FAISS sub-indexes
- Achieved near-zero downtime & high-throughput processing for **>1,000** concurrent user requests by architecting a production-grade deployment on Ubuntu using several Gunicorn workers, load balancing in Nginx, and deduplication in Redis

AI Research Assistant, [Intelligent Manipulation Lab](#), University of Lincoln, Lincoln, UK Mar 2022 – Jan 2023

- Conducted a comprehensive research and developed a novel SMOTE-CNN model for real-time slip classification in robotic manipulation, achieving a 95% reduction in false negatives in a highly imbalanced dataset where non-slip events dominated

Reservoir Performance Simulation Intern, [Schlumberger](#), Port Harcourt, Nigeria

April 2022 – Aug 2022

- Executed maintenance measures and proposed innovative designs, resulting in a 15% efficiency boost and substantial cost reductions
- Conducted over 20 test procedures to assess the integrity of various oil wells, contributing to enhanced operational efficiency

Delegate of Croatia, Model United Nations, Beirut, Lebanon

Mar 2019 – April 2019

- Conducted research on Indigenous cultural heritage and preservation of cultural artifacts and prepared reports for the UNESCO committee

Delegate of Israel, Model United Nations, Buenos Aires, Argentina

Sept 2018 – Oct 2018

- Conducted research on developing policies for reducing global climate change in developing countries

LEADERSHIP EXPERIENCE

Administrative Head, [Luminous Life Foundation](#), Lagos, Nigeria

Sept 2023 – Present

- Led and oversaw a team of 15 to deploy power solutions to underserved homes without electricity for over 50 years in rural Nigeria

Academic Officer, Dept. of Electrical and Electronics Engineering, CU, Ogun, Nigeria

Sept 2022 – Aug 2023

- Served as a liaison for over 500 students and 20 faculty members, ensuring healthy academic welfare
- Coordinated tutorials and curated past questions in collaboration with **TAG** for students across all years
- Organized symposiums with industry leaders to connect Electrical Engineering students to industries

VOLUNTEER WORK & COMMUNITY SERVICE

Co-founder, [Tech for Progress](#), [Isolo, Lagos, Nigeria](#) Jan 2023 – Present

- Initiated and led Isolo's first digital training for underserved youth, equipping 50+ students with essential tech skills

Career Advisor, Tesla's Academic Group (TAG), [CU, Ogun, Nigeria](#) Mar 2022 – Mar 2023

- Guided and supported over 400 students in their academic and career development through periodic webinars

Lead AI Tutor, Google Developer Student Committee (GDSC), [CU, Ogun, Nigeria](#) Jan 2022 – Dec 2022

- Taught and expanded a community of AI enthusiasts from 200 to over 500 through custom curricula
- Led the community to develop AI projects centered around NLP and Computer Vision for Image Detection

Member Assistant Project Manager (National Champions), Enactus, [CU, Nigeria](#) Nov 2019 – Sept 2022

- Led the development of ECOBAG, a heat retention cooler for SMEs, to combat unstable power supply, thereby reducing spoilage
- Assisted in developing PET-City, a plastics-to-bricks scheme, to reduce plastic waste in the environment
- Achieved National Champions status among 18 teams nationwide in 2020 for the impact of PET-City

Tutor, Deeper Life Student Outreach (DLSO), [Ojo, Lagos, Nigeria](#) July 2019 – Aug 2019

- Prepared more than 20 students for college entrance exams while guiding their career choices

SKILLS

Programming Languages: C, C++, Python, PHP, SQL; **Frameworks:** Flask, ROS, Gazebo, OpenCV; **AI & Deep Learning:** Data Analysis, Libraries (Matplotlib, NumPy, PyTorch, sci-kit learn, TensorFlow, seaborn, etc.); **Embedded Systems:** Microcontrollers, Sensors, I2C, SPI, UART, Soldering, KICAD; **CI/CD:** Git, Github; **Soft:** Writing; Teaching; Project, People Management & Time Management; Leadership; Critical Thinking

RELATED PROJECTS

CamAI, Development of an action recognition camera system Mar 2023 – May 2023

- Managed and led a team of two in designing an action recognition system for patient monitoring in healthcare
- Utilized MediaPipe for feature extraction from the camera's video feed and employed an AI ensemble model (Conv-LSTM) to predict the individual's state, while simultaneously displaying the feed on a webpage [\[LINK\]](#)

Quad-X, Development of a Flight Controller for a Quadcopter Jan 2021 – Dec 2021

- Designed and constructed a quadcopter flight controller using an ATMEGA328-PU microcontroller and implemented the PID algorithm and sensor fusion on IMU data for all axis stabilization [\[LINK\]](#)

MilliMeter, Development of a low-cost IoT metering system Mar 2020 – July 2020

- Led a team of 4 to design a low-cost metering system and IoT solution using custom embedded communication protocols and web technologies for an in-state IoT smart metering competition [\[LINK\]](#)

MEDIA APPEARANCES

- Best Graduating Student in Covenant University [\[LINK\]](#) Sept 2023
- 3line's Hackathon [\[LINK\]](#) April 2023

HONORS & AWARDS

- Mastercard Foundation Scholar, Cambridge (69 of 4218) May 2025
- Best Graduating Student (BGS), Covenant University (1 of 1,175) Sept 2023
- BGS in Engineering & EIE, Covenant University (1 of 132) Sept 2023
- Achieved the highest CGPA (4.98/5.0) in the EIE's history at Covenant University Sept 2023
- Dean's Award for Academic Excellence, Covenant University (1 of 386) Sept 2023
- Femi Ademiju's Award for Excellence to the Best Graduating EIE Student Sept 2023
- CUALA's Award for Best Graduating Student in the College of Engineering, Covenant University Sept 2023
- First Runner Up, 3-line Finance Hackathon Aug 2022 – Jan 2023
- First Runner Up, Innovation Hackathon, Covenant University Mar 2022 – May 2022
- \$100 Prize, OMGRobotics Challenge, Goodwall April 2021
- \$250 Prize, My Passion Challenge, Goodwall Feb 2021
- Qualifier, Lagos State Smart Meter Hackathon May 2020 – July 2020
- Winner, Thinkerthon April 2020 – June 2020
- First Runner-Up, Pan-African Hackathon Feb 2020 – May 2020
- Best Student in IGCSE, DLHS, Lagos, Nigeria Aug 2018
- Best Student in Chemistry & Geography Studies, DLHS, Lagos, Nigeria Aug 2018